ADARSH DAMERUPPULA

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PROFESSIONAL SUMMARY

Data Engineer with 4+ years of experience designing, developing, and maintaining scalable ETL/ELT pipelines, distributed data processing systems, and cloud-native data architectures across financial services and enterprise domains. Proficient in Python, SQL, PySpark, Apache Airflow, Snowflake, and AWS for building high-performance data pipelines, data lakehouses, and real-time streaming solutions. Combines deep expertise in data modeling, pipeline optimization, and data quality governance with strong cross-functional collaboration skills to deliver reliable, production-grade data infrastructure that powers analytics and business intelligence at scale.

KEY ACHIEVEMENTS

- Engineered high-scale ETL/ELT pipelines at Bank of America using PySpark, Airflow, and AWS Glue, improving pipeline performance 45% and cutting end-to-end data latency 70% across multi-region AWS environments.
- Architected a Snowflake and Redshift data lakehouse with dbt semantic modeling and Great Expectations quality checks, consolidating 15+ datasets and achieving full SOC 2-compliant data lineage tracking.
- Delivered a cloud-native AWS data platform at S&P Global using Python, SQL, and Airflow, reducing pipeline latency 40%, achieving 90% data availability, and serving 200+ global financial analysts.

CORE COMPETENCIES

ETL/ELT Pipeline Development | Distributed Data Processing | Real-Time & Batch Streaming | Data Lakehouse Architecture | Cloud Data Platforms | Data Modeling & Warehousing | Pipeline Optimization | Data Quality & Governance | Workflow Orchestration | CI/CD & DataOps | Cross-Functional Collaboration | Stakeholder Communication

PROFESSIONAL EXPERIENCE

Senior Data Engineer **01/2024 – Present**

Bank of America, Remote, USA

- Engineered high-scale ETL/ELT pipelines processing billions of financial records using PySpark, Apache Airflow, AWS Glue, and Amazon EMR, improving pipeline performance 45% and reducing processing time 30%.
- Architected a serverless data lakehouse platform using Amazon S3, Snowflake, and Amazon Redshift Spectrum, enabling sub-second query performance and supporting real-time analytics for 50+ operational and finance divisions.
- Developed real-time streaming pipelines using Amazon Kinesis, Apache Kafka, and Flink for high-frequency transactional data ingestion, cutting end-to-end data latency 70% for all downstream analytics consumers.
- Provisioned and managed scalable cloud infrastructure using Terraform, Docker, Kubernetes, and Amazon EKS, ensuring 99.99% pipeline uptime across multi-region AWS environments for all mission-critical data workloads.
- Established CI/CD pipelines using GitHub Actions, AWS CodePipeline, and dbt Cloud with Great Expectations quality validation, ensuring continuous testing and full SOC 2 and GDPR compliance.
- Partnered with risk, compliance, and data science teams to architect regulatory-compliant financial data pipelines, improving data traceability and supporting real-time fraud analytics across 30+ downstream consumers.

Data Engineer **08/2022 – 08/2023**

S&P Global, Bangalore, India

- Developed a cloud-native data platform on AWS, automating ingestion and transformation of high-volume financial data using Python, SQL, and Airflow, reducing pipeline latency 40% across all analytical systems.
- Built and optimized ETL/ELT pipelines using AWS Glue, Lambda, and Redshift to process API and internal system data, improving accessibility for 200+ financial analysts across global business units.
- Implemented data governance protocols using Apache Atlas, AWS IAM, and anonymization techniques to meet GDPR compliance standards, reducing audit remediation time 35% across all production data assets.
- Automated transformation workflows using Python, SQL, and Airflow DAGs, reducing manual data processing efforts 25% and ensuring timely validated delivery for all downstream analytics and reporting teams.

- Designed and maintained data quality monitoring dashboards using Python and SQL, tracking pipeline health across 10+ data sources and reducing data incidents 30% for analytics teams.

Data Analyst **11/2021 – 07/2022**
Genpact, Remote, India

- Analyzed 500K+ daily healthcare claims records using SQL and Python, identifying billing anomalies and coding errors that reduced claim rejection rates 18% across 6 payer systems.
- Built and maintained healthcare data pipelines in Snowflake ingesting EMR and claims data from 4 source systems, improving data completeness 35% for downstream reporting and analytics teams.
- Automated monthly HEDIS and quality measure reporting workflows using Python and SQL scripts, reducing manual preparation time 70% and ensuring on-time delivery for 3 health plan clients.
- Collaborated with clinical and operations stakeholders to document data requirements, map ICD-10 and CPT code logic, and deliver validated datasets supporting population health and utilization analysis.

KEY PROJECTS

Real-Time Fraud Detection Data Pipeline | *Python, Apache Kafka, Spark Streaming, AWS S3, PostgreSQL, Scikit-learn, Flask*

- Designed real-time ingestion pipelines using Kafka and Spark Streaming for continuous processing of transaction data, reducing fraud detection latency from batch-day to under 5 seconds across high-frequency event streams.
- Trained predictive models including Logistic Regression, Random Forest, and XGBoost on historical transaction data, deploying via Flask REST APIs integrated with downstream banking pipelines on PostgreSQL.
- Stored and managed processed transaction data in AWS S3 and PostgreSQL with partitioned schemas, enabling sub-second query performance and supporting downstream fraud reporting for banking analytics teams.

Cloud Data Lakehouse & Analytics Platform | *Python, Snowflake, AWS Redshift, dbt, Airflow, Great Expectations, Terraform*

- Architected a multi-source data lakehouse on Snowflake and AWS Redshift using dbt semantic modeling and Airflow-orchestrated ELT workflows, consolidating 15+ production datasets and enabling same-day analytics delivery.
- Implemented automated data quality checks using Great Expectations and Terraform- provisioned infrastructure, reducing pipeline data discrepancies 45% and achieving full SOC 2-compliant data lineage tracking across all production data assets.
- Built reusable dbt data models and parameterized Airflow DAG templates to standardize transformation logic across 15+ datasets, reducing new pipeline development time 40% for analytical use cases.

EDUCATION

Master of Science in Information Science **01/2024 – 12/2025**
Trine University, United States

Bachelor of Engineering in Electronics & Communication Engineering **08/2019 – 05/2023**
Bharath Institute of Higher Education and Research, India

TECHNICAL SKILLS

Programming & Query Languages: Python (Pandas, NumPy, PySpark, Scikit-learn), SQL, Scala, R, Shell Scripting

Data Engineering & Pipelines: Apache Airflow, AWS Glue, dbt, Apache Kafka, Spark Streaming, Apache NiFi, Databricks

Data Platforms & Warehouses: Snowflake, AWS Redshift, BigQuery, Synapse Analytics, Delta Lake, PostgreSQL, MongoDB

Big Data & Processing: Apache Spark, Hadoop, Hive, Flink, HDFS, Amazon EMR, Amazon Kinesis

Cloud & Infrastructure: AWS (S3, Glue, Redshift, Lambda, EMR, Athena, EKS), Azure (Data Factory, Synapse), GCP (BigQuery, Dataflow)

DevOps & DataOps: Terraform, Docker, Kubernetes, GitHub Actions, AWS CodePipeline, Jenkins, CI/CD, Git

Data Quality & Governance: Great Expectations, Apache Atlas, Data Lineage, GDPR, HIPAA, SOC 2, AWS IAM

Healthcare & Domain: EMR/EHR Systems, ICD-10, CPT Coding, HEDIS, Claims Data, Payer Systems, Population Health, HL7, FHIR

BI & Visualization: Tableau, Power BI, Looker, Amazon QuickSight, dbt Semantic Layer